

Warp Dodgeball — Game Design Spec

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Overview

Warp Dodgeball is team dodgeball where every player carries a portal device: throw a ball into one of your portals and it flies out the other, aimed at an opponent from an angle they never saw coming. The catch is that the physics are honest, so a throw that is too hard can rebound off a wall, come back through your portal, and knock you out.

Core loop: grab a ball → place your two portals → charge and release a throw (straight, curved, or through a portal) → dodge, duck, dive, or catch incoming balls → last team standing wins the round.

Design pillars

- **Readable physics.** Balls arc, bounce, and curve the same way every time, so players can learn bank shots and portal angles.
- **Skill in the release.** Power (how long you hold), spin (curve), and portal placement are the three things a good player masters.
- **Risk and reward.** Every powerful option has a cost: overcharged throws lose accuracy, portal shots can come back at you, and a fumbled catch gets you out.

How to use this spec with Claude Code: build the milestones in order. Each one ends in a playable state with acceptance tests. Numbers in this doc (speeds, timings, sizes) are starting values meant to be tuned, and should live in one config file rather than scattered through code.

Game modes

There are two modes that share every rule and mechanic; they differ only in the arena.

	Classic	Extreme
Arena	One flat floor, no obstacles	Cover blocks, pillars, ladders, raised platforms
Size	18 m × 9 m court	30 m × 16 m arena

	Classic	Extreme
Portal surfaces	Walls and ceiling	Walls, ceiling, obstacle faces, marked floor pads
Balls in play	6	8
Feel	Pure reflex and aim	Hiding, flanking with portals, high ground

Match structure (both modes)

- Teams of 1 to 6 per side (default 4v4). Each team stays on its own half; feet crossing the center line is an out.
- **Opening rush:** balls start lined up on the center line. On the whistle, both teams sprint for them. A ball grabbed in the rush must be carried back behind your attack line (3 m from center) before it can be thrown.
- A round ends when one team has no players left, or after 3 minutes. On timeout, the team with more players standing wins; a tie goes to sudden death (one ball, no portals).
- A match is best of 5 rounds.
- Eliminated players wait on their team's sideline in catch order: when a teammate makes a catch, the first player in line returns.

Player movement and controls

Every evasive move trades safety for something: lower hitbox for less speed, a long dive for a vulnerable recovery. The player is a capsule 1.8 m tall and 0.5 m wide; the hitbox shrinks with posture.

Move	Behavior	Hitbox height	Speed	Can throw?
Walk	Default movement	1.8 m	4.5 m/s	Yes
Sprint	Hold to run; drains nothing, but you can't charge a throw	1.8 m	6.5 m/s	No
Jump	1.2 m apex; limited air control (30%)	1.8 m	Keeps momentum	Yes, mid-air
Crouch	Hold to stay low; works with cover in Extreme	1.1 m	2.0 m/s	Yes

Move	Behavior	Hitbox height	Speed	Can throw?
Squat (quick duck)	Tap for an instant full-depth drop lasting 0.4 s, then auto-stand; 0.6 s cooldown	0.9 m	Stationary	No
Dive	Lunge 3 m in the input direction (any of 8), land prone, 0.8 s to recover	0.5 m (prone)	Burst	No, until recovered
Ladder climb	Snap to ladder on contact + forward input; mantle onto the platform at the top; jump to drop off	1.8 m	2.0 m/s	No

Input map

Action	Keyboard and mouse	Gamepad
Move / look	WASD / mouse	Left stick / right stick
Sprint	Left Shift	L3 (click left stick)
Jump	Space	A
Crouch (hold)	Left Ctrl	B (hold)
Squat (tap)	C	R3 (click right stick)
Dive	Q + direction	X + direction
Throw (hold to charge, release to throw)	Left mouse	Right trigger
Cancel throw	Right mouse	D-pad down
Curve left / right	Hold Z / X at release, or flick mouse sideways	Right stick sideways at release
Place portal A / B	E / R	LB / RB
Catch (quick-time prompt)	F	Left trigger

Balls are picked up automatically by walking over a resting ball on your side. A player can hold one ball at a time.

Throwing

Throw power is set by how long the throw button is held: a tap lobs, a full charge fires a 26 m/s fastball, and holding too long makes your aim shaky.

Charge

- Charge fills from 0 to 1 over 1.2 s. A meter around the crosshair shows it.
- Release speed = $8 \text{ m/s} + 18 \text{ m/s} \times \text{charge}^2$, so short holds stay gentle and the last part of the charge adds the most speed.
- Weak throws loft upward so they arc: launch angle = camera pitch + 10° at zero charge, falling to $+2^\circ$ at full charge.
- **Overcharge:** after 2.0 s held at full, aim spread grows by 2° per second (up to 6°). This stops players camping on a loaded throw.
- Charging slows movement to 70%. The release animation takes 0.15 s; the ball leaves the hand at the end of it.

Curve balls

- Holding curve left/right (or flicking sideways) at the moment of release adds sidespin, scaled by charge, up to 30 rad/s.
- Spin bends the flight path through a Magnus force: $F = k \times (\text{spin} \times \text{velocity})$. Tune k so a full-curve fastball drifts about 3 m sideways over 20 m.
- Spin decays 15% per second and is cut in half by each bounce.
- Spin is kept when the ball passes through a portal (rotated with it), so curved portal shots are possible.
- Optional later: vertical flicks for topspin (dips) and backspin (floats).

Practice aid: in practice mode only, show the predicted arc for the first 0.5 s of flight. Never in matches.

Catching

Catching is a timed button press: hit it inside the window and the thrower is out and a teammate comes back; miss the timing and you fumble and are out yourself.

When the prompt appears

- A live enemy ball is predicted to hit you within 0.6 s, and
- it is coming from within 100° of the direction you're facing, and
- you are standing, crouching, or jumping (not diving, recovering, squatting, or on a ladder), and
- your hands are empty. Holding a ball, you can **block** instead: the ball is deflected and goes dead. If the hit knocks your ball out of your hands, you are out.

Only the most urgent threat gets a prompt. The prompt is a shrinking ring around the incoming ball with the catch button icon.

The window

Ball speed	Catch window
Under 12 m/s	0.35 s
12 to 20 m/s	0.25 s
Over 20 m/s	0.15 s

The window is centered on the predicted moment of contact.

Outcomes

- **Pressed inside the window:** catch. The thrower is out, your first eliminated teammate returns, and you now hold the ball.
- **Pressed outside the window (early or late):** fumble. The ball pops off you and you are out.
- **Not pressed:** an ordinary hit, so you are out unless you dodged.

Because pressing early is as bad as getting hit, players must decide between catching and dodging, which is the core tension of dodgeball.

Portal system

Each player owns one linked pair of portals (A and B). A ball that enters one exits the other at the same speed, turned to match the exit portal's facing. Only balls travel through portals; players cannot, which keeps the center-line rule meaningful.

Placement

- Pressing portal A or B fires an instant ray from the camera. The portal is placed if the hit surface is portal-able and a 1.0 m × 1.6 m oval fits flat on it.
- Portal-able surfaces are light gray panels; everything else is dark and rejects portals. This lets level design control which angles exist.
- Portals can go anywhere valid, including the opponent's side of the arena.
- Placing A again moves your existing A. Cooldown: 1.0 s per portal.
- Your portals are tinted in your team color, light for A and dark for B, with your player number on the rim. Opponents see them too.
- When you are eliminated, your portals close. A ball in mid-transit finishes its trip.
- Portals are only active when both ends of the pair exist.

Transit (the math Claude Code needs)

- Trigger when the ball's center crosses the portal plane inside the oval.
- New position = exit transform × 180° turn about the portal's up axis × inverse of entry transform × old position.
- Velocity and spin are rotated by the same rotation. Speed is unchanged.
- For 0.1 s after exiting, the ball ignores collisions with the wall behind the exit portal.
- Anyone's ball can go through anyone's portals. That includes the enemy's, so a portal placed carelessly can be used against you.
- Portal views are rendered so players can see through them (a second camera per portal rendering to a texture). Recursive views are capped at 1 level.

Rebound out (the risk rule)

A live ball that hits its own thrower after passing through any portal, or after bouncing off a wall, gets the thrower out. In practice this happens when a player throws too hard through a portal that faces a nearby wall: the ball keeps enough speed to bounce back through the portal pair toward them. Lighter throws lose more energy and die on the floor first, so power is a choice, not a default.

To stop instant self-hits on release, the thrower is immune to their own ball for 0.3 s after release unless it has already been through a portal.

Ball physics and elimination rules

A ball is live from release until it touches the ground; walls, ceilings, and obstacles do not kill it, which is what makes bank shots and portal rebounds possible.

Ball properties

Property	Value
Diameter	0.21 m (standard 8.25 in dodgeball)
Mass	0.25 kg
Gravity	9.81 m/s ²
Air drag	Quadratic, drag coefficient 0.47
Bounciness off floor	0.60
Bounciness off walls and obstacles	0.70
Surface friction	0.5, with rolling damping so balls come to rest
Physics rate	120 Hz, with continuous collision detection (a 26 m/s ball moves 0.2 m per tick)

Ball states

1. **Held:** in a player's hands.
2. **Live:** thrown, and tagged with the thrower and team. Can get players out.
3. **Dead:** harmless. A ball becomes dead when it touches the ground, hits a player, is caught, or is blocked. "Ground" means any walkable surface: the floor and the tops of platforms and cover blocks (any surface tilted less than 45°). Entering a floor portal does not count as touching the ground.

How a player gets out

1. Hit by a live enemy ball anywhere on the body.
2. Hit by their own live ball after it went through a portal or bounced off a wall (rebound out).
3. Their throw is caught by an opponent.
4. They fumble a catch.
5. A hit knocks the ball out of their hands while blocking.
6. Their feet cross the center line.

A ball only gets one player out per flight; after the first hit it goes dead. Friendly fire is off by default (teammates' balls pass harmlessly), with a lobby setting to turn it on. A lobby setting can also make headshots not count.

Arenas

Both arenas are mirror-symmetric so neither team has an advantage, and both are enclosed so balls stay in play.

Classic court

- Floor: 18 m × 9 m, with a center line and an attack line 3 m back on each side. A 2 m run-off strip surrounds the court.
- Walls 6 m high on all sides; flat ceiling at 8 m.
- Portal-able panels: the full end wall behind each team, the upper half (3 to 6 m) of both side walls, and the whole ceiling. The floor is not portal-able in Classic.
- Nothing else. The mode is about pure aim, timing, and portal angles.

Extreme arena

Floor 30 m × 16 m, walls 8 m, ceiling 10 m. Each half contains the same set of pieces, mirrored:

Element	Count per side	Size	Purpose	Portal-able
Low cover block	3	2 m wide × 1.0 m tall	Crouch behind for cover	Back face only
Tall pillar	2	1.5 m × 1.5 m × 3 m	Full cover, blocks sight lines	One side face each
Raised platform (vantage nest)	2	3 m × 3 m deck at 3.5 m, 1 m railing	High ground, sees over cover	Railing faces no; wall behind yes
Ladder	2 (one per nest)	3.5 m	Only way up to a nest	No
Angled deflector wall	2	3 m wide × 2.5 m tall, set at 30°	Bank shots around cover	No
Floor portal pad	2	2 m circle	Lets balls drop in and launch up from below	Yes

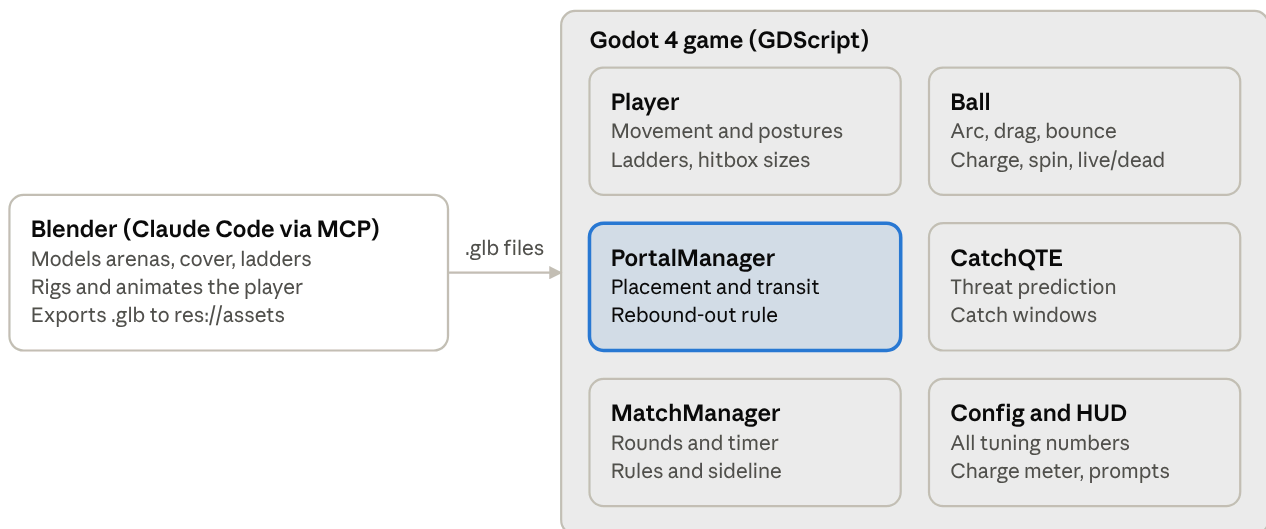
Design notes:

- Nests are powerful (you can see over all cover), but climbing the ladder takes about 2 s during which you can't throw, catch, or dodge.
- Cover is placed so that every hiding spot can be reached by at least one portal angle. Hiding is never fully safe.
- Nest decks count as ground: a ball that lands on a deck goes dead.

Technical architecture and Blender pipeline

Blender is the asset tool and Godot 4 is the game engine. Blender no longer has a built-in game engine (it was removed in version 2.80), so Claude Code should model in Blender and run the game in Godot, which is free, text-based (easy for Claude Code to edit), and imports Blender files directly.

Claude Code models assets in Blender; Godot runs the game



build architecture · Blender assets into six Godot systems

PortalManager is highlighted because it is the riskiest system: transit math and see-through rendering need the most testing.

Blender conventions

- Export each asset as .glb into the Godot project's assets folder; Godot re-imports on change.
- Name collision meshes with a `-col` suffix so Godot builds collision shapes automatically.

- Mark portal-able faces with a material named `portal_surface` ; the game reads this material to allow placement.
- Keep greybox and final art as separate files, so gameplay work never waits on art.

Engine notes

- Player: a CharacterBody3D with a capsule collider resized per posture.
- Ball: a RigidBody3D with continuous collision on, with drag and Magnus force added each physics tick.
- Portals: an Area3D trigger plus a SubViewport camera rendering the view through the portal.
- Match flow: a state machine (warmup → opening rush → live → round end → match end).
- Networking, if wanted later: server-authoritative, so the server decides every hit and catch.

Build milestones for Claude Code

Build in this order; portals come after the dodgeball itself feels good, because portal shots are only fun if the throwing underneath them is solid.

1. **Project setup and greybox.** Godot project, folder layout, config file for all tuning numbers. Model the Classic court greybox in Blender and import it.
 - Test: the court loads, the player camera spawns on the correct half.
2. **Player controller.** Walk, sprint, jump, crouch, squat, dive, with the hitbox sizes from the movement table.
 - Test: a debug overlay shows hitbox height changing for each posture; dive recovery locks input for 0.8 s.
3. **Ball physics and throwing.** Pick-up, hold-to-charge, arc, drag, bounce, live/dead states, continuous collision.
 - Test: a zero-charge throw lands in a visible arc; a full throw at 26 m/s never tunnels through a wall in 1,000 automated throws.
4. **Curve balls.** Spin input and Magnus force.
 - Test: a full-curve fastball drifts 2.5 to 3.5 m sideways over 20 m.
5. **Rules, match flow, and bots.** Eliminations, center line, opening rush, rounds, sideline queue, simple AI opponents (pick up, aim, throw, dodge randomly).
 - Test: a 4v4 bot-only match completes 5 rounds with no stuck states.

6. **Catching.** Prompt, speed-based windows, catch/fumble/block outcomes, teammate return.
 - Test: scripted balls at 10, 16, and 24 m/s produce windows of 0.35, 0.25, and 0.15 s.
7. **Portals.** Placement rules, transit math, see-through rendering, rebound-out rule.
 - Test: a ball entering at 20 m/s exits at 20 m/s in the correct direction for floor, wall, and ceiling portal combinations; a hard throw into a portal facing a near wall gets the thrower out.
8. **Extreme arena.** Model cover, pillars, nests, ladders, deflectors, floor pads in Blender; ladder climbing and mantling.
 - Test: every cover spot can be hit by at least one portal angle (automated sweep).
9. **Polish.** Art pass on models and the player character in Blender, sounds, hit effects, menus, mode select, tuning pass.

Open questions

These have a working default in the spec; change them before Claude Code starts the matching milestone.

- ☐ **Players:** local vs. bots only, split-screen, or online multiplayer? Default: single player against bots first, networking later.
- ☐ **Camera:** first-person (best for aiming portals) or third-person (best for seeing dives)? Default: first-person.
- ☐ **"Squat":** spec treats it as a quick duck distinct from crouch. Confirm that's what you meant.
- ☐ **Portal ownership:** one pair per player (default) or one shared pair per team?
- ☐ **Players through portals:** off by default. Allowing it would need a rule for crossing the center line.
- ☐ **Art style:** low-poly stylized, or something closer to realistic?